OCR APIs Doc Classifier

1) https://docclassifier1-dev.radflow360.com:4000/ocr_process This API performs OCR processing on documents such as cheques (in PDF format) and photo IDs (in image format). The API takes documents where each document contains a base64-encoded string and its type (`photo_id` or `cheque`). For each document, it first decodes and saves the file temporarily, auto-rotates it if needed, and then performs OCR (text extraction). It primarily uses **AWS Textract** for text extraction, and if that fails, it falls back to a secondary method using **Pytesseract**. If the document is a **photo ID**, it extracts important details like first name, last name, date of birth (DOB), address, ID number, and other personal information. If the document is a **cheque**, it extracts fields like first name, last name, DOB, DOI (Date of Issue), and searches for a patient ID (starting with "PRE"). If the patient ID is not found directly in the text, the system attempts to find the patient by matching the extracted first name, last name, and DOB or DOI by calling a **.NET-side developed API**. After extracting the necessary information, the document is updated with these details under the `"info"` key, the processed file is re-encoded into base64 format, and everything is returned. If any error occurs during the process, the API responds with a standard error message. Temporary files created during processing are deleted afterward to keep the server clean.

    a) PAYLOAD: {
    "data": [
        {
            "docType": "cheque", # docType should be “cheque” or “photo_id”
            "encoded_string": “<cheque_or_photo_ID_encoded_string>”
        }
    ]
    }

    b) RESPONSE: {
        "data": [
            {
                "docType": "cheque",
                "encoded_string": “<cheque_encoded_string>”,
                "info": {
                    "dob": [],
                    "doi": [],
                    "first_name": “”,
                    "last_name": “”,
                    "patient_id": []
                }
            },
            {
                "docType": "photo_id",

```
            "encoded_string":  “<photo_id_encoded_string>”,
            "info": {
                "address": "",
                "dob": [],
                "expiry_date": [],
                "eyes": "",
                "first_name": "",
                "hair": "",
                "height": "",
                "id_number": "",
                "last_name": "",
                "patient_id": [],
                "sex": "",
                "weight": ""
            }
        }
    ],
    "message": "Success"
}
```


2) https://docclassifier1-dev.radflow360.com:4000/pdf_ocr_process This API processes PDF documents by extracting important information from them. It accepts documents where each document contains a base64-encoded PDF file. For each document, it decodes the base64 string, reads the PDF in memory, and extracts text from each page. To speed up the extraction, it uses **multi-threading** (via `ThreadPoolExecutor`) and processes multiple pages in parallel. After collecting all the text, it searches for specific patterns such as the **patient ID** (starting with "PRE") and uses a custom OCR method to pull out structured information. The extracted data is added under the `"info"` key and returned with fields like `body_part`, `cpt`, `dob`, `first_name`, `last_name`, `patient_id`, and `study_type`. If a patient ID is found, it is included in the `"patient_id"` field; otherwise, that field remains empty. All temporary resources are cleaned up after processing. If any error occurs, the API responds with an appropriate error message. This API is particularly optimized for handling multi-page PDFs efficiently and returning structured data for further processing.

    a) PAYLOAD: {
    "data": [
        {
            "encoded_string": “<encoded_string>”
        }
    ]
}

    b) RESPONSE: {
        "data": [
            {

```
            "encoded_string":  “<encoded_string>”,
            "info": {
                "body_part": "",
                "cpt": "",
                "dob": "",
                "first_name": "",
                "last_name": "",
                "patient_id": "",
                "study_type": ""
            }
        }
    ],
    "message": "Success"
}
```


3) https://docclassifier1-dev.radflow360.com:4000/all_doc_ocr_process This API is built to handle both image files and PDF documents for OCR-based text extraction. It accepts a list of documents where each document contains a base64-encoded string. For each document, it first checks if the content is an image (PNG or JPEG); if so, it automatically converts the image into a PDF format for easier text extraction. It then reads the PDF in memory and extracts text from each page using **multi-threading** (via `ThreadPoolExecutor`) to process pages in parallel for better performance. After extracting all the text, it looks for specific patterns such as **patient IDs** (starting with "PRE") and **accession numbers** (starting with "RAM"). It uses a custom method to also identify **study types** based on the extracted text and any found accession numbers. The API then organizes the extracted details into an `"info"` field with three keys: `accession_nums` (list of found accession numbers), `patient_id` (first found patient ID or empty if none), and `study_type` (list of extracted study types). All processed data is returned to the client. If any errors occur during processing, the API responds with an appropriate error message. This API is particularly useful for handling mixed uploads of scanned images and PDFs in healthcare and document management workflows.

    a) PAYLOAD:

```
{
    "data": [
        {
            "encoded_string": “<encoded_string>”
        }
    ]
}
```


    b) RESPONSE:

```
{
    "data": [
        {
            "encoded_string":  “<encoded_string>”,
            "info": {
                "accession_nums": [],
```

```
            "patient_id": "",
            "study_type": []
        }
    }
  ],
  "message": "Success"
}
```